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Learning to inversely design acoustic metamaterials for enhanced performance

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Elastic metamaterials are popularly sought to realize numerous special functions such as vibration control and wave manipulation among which sound absorption is a typical task fulfilled by acoustic metamaterials. Inverse designing metamaterials with machine learning approaches has been under the spotlight thanks to the data-driven experience-free advantages and become one of the important design paradigms. Nevertheless, the existing works mostly concentrate on validating the reproduction accuracy of the neural networks on trained data and very few have explored their ability on designing for enhanced properties. To this end, our work studies the competence of the proposed inverse design framework in enhancing the acoustic performance of a three-dimensional mixed-size cavity-based waterborne sound absorptive metamaterial. With forward and inverse networks in the framework, the target sound absorption spectra (100-10000 Hz) are taken as inputs into the inverse network during training and a corresponding structure is output with the best matching spectra which is subsequently fed into the forward network for acoustic property evaluation and loss calculation. The trained forward network is shown to possess excellent generalization capabilities by highly accurately predicting for structures with "unseen" beyond-range parameters compared to the training set. Most importantly, the inverse network is delightfully capable of spontaneously adopting beyond-range structural parameters to ensure meeting the acoustic target whose mean sound absorption coefficient is higher than any of the data in the training set, hence inverse designing for enhanced performance. The inverse design accuracy is dramatically improved from only 9.2% of mean squared errors being < 0.0001 to 99.6% with beyond-range exploration. A case study is presented to demonstrate the significant difference beyond-range exploration makes for inverse designing aiming at enhanced performance. It is hoped that this work will serve as an inspiration for the design and optimization of elastic metamaterials with enhanced performance for future work.

Acoustic metamaterials, Inverse design, Machine learning, Waterborne sound absorption

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1. Introduction

Acoustic metamaterials, as a typical kind of elastic metamaterials consisting of self-repetitive artificial substructures referred to as unit cells, feature excellent capabilities of acoustic wave manipulation for sound absorption or insulation [1-5]. The designing of the unit cells plays a vital role in determining the acoustic properties of the metamaterials and is normally carried out based on physics-based models and optimisation algorithms (such as genetic algorithms [6]), which are heavily dependent on expert domain knowledge and numerical simulation methods. Additionally, iterative optimisation with numerical simulation is usually inefficient and expensive especially when it comes to three-dimensional (3D) models where axisymmetric boundary conditions are inapplicable. To this end, machine learning is under the spotlight and is expected to empower the design of

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metamaterials as a new paradigm [7] with the advantages in data-driven learning of hidden complex non-linear linkage between structure and properties [8-11]. Among all the designing frameworks of metamaterials with machine learning, inverse design is the most target-driven and task-oriented approach. The general workflow is to feed the on-demand properties to the model which simultaneously outputs the feasible design of the metamaterial which meets the input demands. An inverse design architecture normally contains an inverse net to design under the demand for properties and a forward net to evaluate the property of the metamaterial given by the inverse net.

Researchers have been eagerly exploring the applications of inverse design with machine learning models on both optical [12-17] and elastic [18-21] metamaterials. Gurbuz et al. [20] harassed conditional general adversarial networks (GANs) to inversely design metamaterials for sound insulation tasks with binary images representing the metamaterials. Hou et al. proposed an inverse model that took the frequency domain response of a metamaterial absorber as the input and designed the corresponding geometry and material parameters [22]. Oddiraju et al. [18] employed invertible neural networks (INN) for fast inverse design of locally resonant elastic metamaterials for vibration control. Tian et al. [19] combined molecular dynamics for data preparation and cycle-GAN to inversely design two-dimensional (2D) metamaterials with tunable deformation-dependant Poisson's ratio. A trained forward INN aimed to predict the bandgap bounds from the unit cell, while an inverse was designed to obtain a specified bandgap. Ma et al. [23] modelled the inverse design problem in a probabilistically generative manner. They specially designed the representation of metamaterials with variational auto-encoder to obtain a 20-dimentional latent space whose latent variables were used together with the target reflection spectra as the input for the generative model. A generative model with a generator, a simulator, and a critic was proposed for the inverse design of optical metasurfaces with on-demand spectra [24]. The model was proven capable of accurately inversely designing a structure from the training set with its corresponding transmittance spectra as the input, which was essentially inversely reproducing a datum that had been "seen" by the forward model to examine the inverse designing accuracy of the model.

All the works above have successfully demonstrated the effectiveness and reliability of machine learning models in inversely designing metamaterials when the target properties are reasonably within reach in the sense that they already exist in the training set and have already been "learned" by the model. In the meantime, whether a machine learning model can predict or design beyond the range considered in the input database has become one of the key questions researchers are seeking answers for [25]. As machine learning algorithms are advantageous in capturing the hidden relationship between the unit cell structures and the metamaterial properties, it possesses a great potential to exploit the learned knowledge and design beyond the range considered for performance that exceeds the existing ones. Pushing the inverse model to design for significantly enhanced properties compared to the original training data is the main purpose of our study.

In this paper, a neural-network-based inverse-designed framework is proposed for 3D mixed-size cavity-based acoustic metamaterials for broadband (100-10000 Hz) waterborne sound absorption. The inverse design is shown capable of outputting structures beyond the range considered as well as with enhanced sound absorption performance compared to the training data. The outline of the paper is as follows. The acquirement of the training and testing datasets is first introduced including the details of the metamaterial and the finite element method (FEM) simulation, followed by the workflow and training process of the inverse design framework. Results are subsequently discussed regarding the accuracy of the forward network prediction on structural parameters beyond the range considered in the training set and the capability of the inverse network to design metamaterials for enhanced sound absorption performance with beyond-range structural parameters. In the end, conclusions are drawn and future work is briefly discussed. The codes and datasets are available at https://github.com/h-zhang00/Learning-to-Inversely-Design-Acoustic-Metamaterials-for-Enhanced-Performance.

2. Dataset

The sound absorption property prediction and the structural inverse design of 3D mixed-size cavity-based acoustic metamaterials for waterborne sound absorption (referred to as "absorber" in the following discussion) are investigated. The reason for choosing this type of metamaterial is that the rotational symmetry usually adopted to simplify the numerical model is not applicable anymore. Consequently, its iterative optimization with the traditional approaches is unfavourably computationally expensive as well as dramatically tedious and inefficient. Machine learning models, on the other hand, are extremely fast in property prediction once trained and enjoy the end-to-end advantages of sparing the iterative process. Therefore, machine learning approaches are of more significance in the case of dealing with 3D models where rotational symmetry is not available.

The configuration of the mixed-size waterborne acoustic absorber is shown in Fig. 1(a). It is made of a rubber layer embedded with a doubly periodic array of cone-cylindrical combined voids. The absorber is covered by a half-infinite water domain on the top with a perfectly matched layer

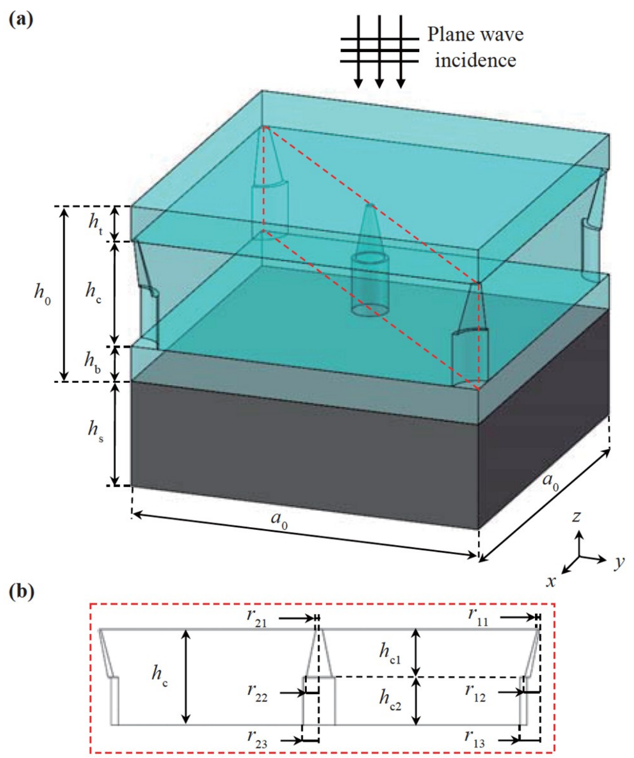

Figure 1 Schematic diagram of (a) the waterborne acoustic absorber and (b) the cross-section of the voided layer.

(PML) to simulate the anechoic termination of outgoing waves and backed with a steel plate on the bottom. A free boundary condition is applied on the undersurface of the steel backing and the interface between the rubber matrix and voids. To guarantee the periodicity in the x- and y-directions, a periodic boundary condition is adopted on the side walls of the absorber. Furthermore, considering the interaction between different media, an acoustic-structure boundary condition is implemented on the interface between the fluid domain and the solid domain. A plane sound wave impinges on the absorber from the water at normal incidence with $p_i = p_0 \exp(\mathrm{i}k_0 z)$ where the time-harmonic factor $\exp(\mathrm{i}\omega t)$ is omitted and p_0 and k_0 are the amplitude of the acoustic pressure and the wavenumber of water. In addition, the lattice constant is a_0 and the heights of the cover plate, cavity plate, baseplate and steel plate are h_t, h_c, h_b, and h_s, respectively, where the total height of the rubber layer satisfies the condition of "$h_0 = h_t + h_c + h_b$". The combined cavities in the four corners are the same, different from the cavities in the middle of the matrix. The radii of the upper base and lower base of the cone, and the cylinder in the corner are r_{11}, r_{12}, r_{13}, respectively, and the corresponding radii of the middle cavities are r_{21}, r_{22}, r_{23}, respectively, as depicted in Fig. 1(b). The heights of cones and cylinders are h_{c1} and h_{c2}, respectively, meeting the condition of "$h_c = h_{c1} + h_{c2}$". Overall, the structure of the waterborne acoustic absorber is characterised by 10 structural parameters to be inversely designed.

The range of variation for structural parameters is listed in Table 1. Note that there are two sets of ranges, i.e., range1 and range2. Generally, range2 is 50% larger than the range1 except for the lattice constant a_0. The aim of having two sets of ranges is to examine the neural network's generalisation capability as in whether it can correctly evaluate the property of a structure of unseen sizes or whether it can inversely design a metamaterial of beyond-range structural parameters necessary to meet the on-demand sound absorption target. This will be further discussed to explain the difference between the three datasets used in this work.

The absorption response of the absorber in concern is from 100 Hz to 10000 Hz and is discretised into 109 frequency points with an interval of 50 Hz between 100 Hz and 1000 Hz and 100 Hz between 1100 Hz and 10000 Hz. FEM via COMSOL software is used to calculate the corresponding SACs for 109 frequency points. The acoustic parameters of materials in the FEM simulation are shown in Table 2.

Datasets of waterborne acoustic absorbers are created as

$$\mathbb{D} = \left\{ \left(\mathcal{S}_i^*, \mathcal{A}_i^* \right), \quad i = 1, 2, \ldots, n \right\}, \tag{1}$$

$\mathbb{D}$ includes multiple sets of structure-property pairs where $\mathcal{S}_i^*$ is the assembly of 10 independent structural parameters and $\mathcal{A}_i^*$ is the sound absorption spectra as 109 sound absorption coefficients (SACs). An asterisk is added here to specify that the data are ground truths prepared with FEM simulation.

Specifically, there are one training set $\mathbb{D}$ and two testing sets ($\mathbb{D}_\mathrm{I}$ and $\mathbb{D}_\mathrm{II}$) for different testing purposes. The key information is listed in Table 3. $\mathbb{D}$ contains 20000 pairs of $\mathcal{S}^*$-$\mathcal{A}^*$ data whose average SACs are all below 0.82 and structural parameters are within range1. Testing set $\mathbb{D}_\mathrm{I}$ has 1000 pairs of $\mathcal{S}^*$-$\mathcal{A}^*$ data but with structural parameters following range2, which is the main difference between $\mathbb{D}$ and $\mathbb{D}_\mathrm{I}$. Testing set $\mathbb{D}_\mathrm{II}$ consists of 500 data of only sound absorption spectra and is used as the target for inverse de-

Table 1 Range of variation for structural parameters

	s1	s2	s3	s4	s5	s6	s7	s8	s9	s10
	a_0	r_{12}	h_{c1}	r_{13}	h_{c2}	r_{22}	r_{23}	h_t	r_{11}	r_{21}
range1-min	50	2	2	2	2	2	2	5	1	1
range1-max	100	10	19	10	19	10	10	10	10	10
range2-min	50	1	1	1	1	1	1	2.5	0.5	0.5
range2-max	100	15	28.5	15	28.5	15	15	15	15	15

Table 2 Acoustic parameters of the materials in FEM simulation

Materials	Density (kg/m^3)	Longitudinal-wave speed (m/s)	Shear-wave speed (m/s)	Loss factor
Water	1000	1500	–	–
Air	1.29	340	–	–
Rubber	1100	1500	100	0.8
Steel	7850	5780	3220	–

Table 3 Key information of three datasets

	Name	Size	Data type	Structural range	Average SACs
$\mathbb{D}$	Training set	20000	$\mathcal{S}^*$-$\mathcal{A}^*$	range1	< 0.82
$\mathbb{D}_{\mathrm{I}}$	Testing set I	1000	$\mathcal{S}^*$-$\mathcal{A}^*$	range2	–
$\mathbb{D}_{\mathrm{II}}$	Testing set II	500	$\mathcal{A}^*$	–	≥ 0.82

sign. Note that the average SACs of these 500 spectra are all over 0.82 as contrary to that in $\mathbb{D}$, meaning that we will ask the inverse network to design for acoustic properties exceeding all the absorbers it is trained with, hence designing for exceeding performance.

3. Inverse design framework

The inverse design framework is composed of two parts—an inverse network $\mathbb{I}$ and a forward network $\mathbb{F}$, as shown in Fig. 2.

Both the inverse and the forward networks are multi-layer artificial neuron networks (ANNs) containing fully connected layers, intermediate non-linear activation layers, and an output layer. The squared L2 norm is adopted as the loss function for training the networks, which measures the mean squared error (MSE) between the input and the output as described by the following equation:

$$\mathcal{L}(\mathbf{x}, \mathbf{x}^*) = \|\mathbf{x} - \mathbf{x}^*\|^2 = \frac{1}{m}\sum_{j=1}^{m}\left(x_j - x_j^*\right)^2, \tag{2}$$

where m is 109 for frequency points characterizing the sound absorption spectrum $\mathcal{A}$ in our case. $\mathbb{I}$ and $\mathbb{F}$ are trained separately.

$\mathbb{F}$ is firstly trained with $\mathbb{D}$ and acts as a substitute for the FEM simulation to evaluate the sound absorption performance of an inputting metamaterial during the training of $\mathbb{I}$. It is fed with the 10 structural parameters representing the structure of the metamaterial. Five intermediate fully connected layers are sequentially arranged (containing 128, 64, 64, 32, and 32 neurons, respectively) with leaky rectified linear unit (leaky Relu) activation layers in between followed by an output layer with 109 neurons and finally a Sigmoid layer to output 109 SACs corresponding to 109 frequency points. The learning rate is 1×10^{-3} and the batch size is 200. The MSE loss gives a direct comparison between the ground truth sound absorption performance and the one predicted by $\mathbb{F}$. Once the forward network $\mathbb{F}$ is well-trained,

$$\min_{\kappa}\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\kappa}\left[\mathbb{F}(\mathcal{S}_i), \mathcal{A}_i^*\right] \to \mathbb{F}. \tag{3}$$

The inverse network $\mathbb{I}$ can effectively and efficiently reversely design the structure of the metamaterials and ensure the approximation of their sound absorption properties to the target ones. The inverse network resembles the forward one except that there are 6 intermediate layers with 1024, 512, 256, 128, 64, and 32 neurons, respectively. The learning rate is 5×10^{-4} and the batch size is 200. AsidFe from some differences in hyperparameters (e.g., the size of fully connected layers and the depth of the model) between $\mathbb{F}$ and $\mathbb{I}$, the major difference lies in the calculation of the loss. It is highly possible that various structures share almost the same sound absorption performance, and this is the famously known one-to-many property-structure mapping issue inherently featured by inverse design [21]. Therefore, it is unwise to approximate the output structural parameters $\mathcal{S}$ by $\mathbb{I}(\mathcal{A})$ to the corresponding structure $\mathcal{S}^*$ of $\mathcal{A}$ in the training set $\mathbb{D}$, which would only overfit $\mathbb{I}$ to the training data and limit its capability in designing for enhanced variety and "imagination". Instead, the more important accuracy to be measured is whether the inversely designed metamaterial $\mathcal{S} = \mathbb{I}(\mathcal{A}_i)$'s sound absorption performance $\overline{\mathbb{F}}[\mathcal{S}]$ meets the target of sound absorption performance $\mathcal{A}_i^*$, the MSE of which is the loss for training $\mathbb{I}$. Note that $\overline{\mathbb{F}}[\mathcal{S}]$ illustrates that

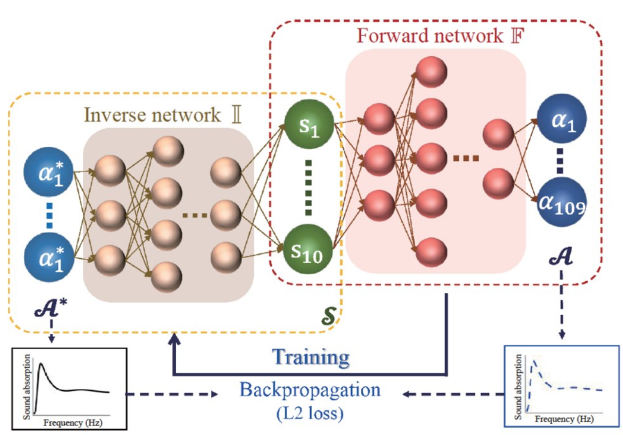

Figure 2 Architecture of the inverse design framework.

the sound absorption performance is calculated by the forward network $\overline{\mathbb{F}}$ trained beforehand and adding a bar above here means the weights of the forward network are fixed during the training of $\mathbb{I}$. Similarly, $\overline{\mathbb{I}}$ means a trained inverse network.

$$\min_{\gamma}\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\gamma}\left\{\overline{\mathbb{F}}[\mathbb{I}(\mathcal{A}_i)],\mathcal{A}_i^*\right\}\rightarrow\mathbb{I}. \tag{4}$$

All the data are normalized and 10-fold cross validation is used during the training [26] to avoid overfitting and reduce the dependence of the network performance on the train-evaluation division of the datasets.

Since both $\overline{\mathbb{F}}$ and $\overline{\mathbb{I}}$ are trained with $\mathbb{D}$, they have only "encountered" structural parameters within range1 and absorbers whose average sound absorption is below 0.82. The testing of $\overline{\mathbb{F}}$ is carried out with $\mathbb{D}_{\mathrm{I}}$ whose data feature structural parameters following range2 (50% larger than range1). Therefore, some data in $\mathbb{D}_{\mathrm{I}}$ are entirely new to $\overline{\mathbb{F}}$. The expectation is the output of $\overline{\mathbb{F}}$ can give an accurate prediction for structural parameters beyond range1 considered in the training set $\mathbb{D}$ [25].

$$\mathbb{T}^{fwd}=\left\{\mathcal{A}_i^{fwd}=\overline{\mathbb{F}}(\mathcal{S}_i)\right\},\ \mathcal{S}_i\in\mathbb{D}_{\mathrm{I}},\ i=1,2,\ldots,1000. \tag{5}$$

This is to verify whether $\overline{\mathbb{F}}$ has learned the essential underlying non-linear relationship between the structure and the sound absorption properties and is capable of generalising and expanding this knowledge to predict for unseen structures. The inversely designed absorbers are obtained by $\overline{\mathbb{I}}$ with the 500 sound absorption spectra whose average SACs are over 0.82 in $\mathbb{D}_{\mathrm{II}}$ as the targets, as a comparison to the average SACs below 0.82 in the training set $\mathbb{D}$.

$$\mathbb{T}^{inv}=\left\{\mathcal{S}_i^{inv}=\overline{\mathbb{I}}(\mathcal{A}_i)\right\},\ \mathcal{A}_i\in\mathbb{D}_{\mathrm{II}},\ i=1,2,\ldots,500. \tag{6}$$

Hence, we are essentially asking the inverse network to design for targets beyond the training data. The expectation is that the inverse network's design can break the limits of the structural parameter range in the training set to meet the target spectra.

4. Results and discussion

4.1 Forward network $\overline{\mathbb{F}}$

The training of the forward network took roughly 2 hours to obtain $\overline{\mathbb{F}}$. Once trained, the time needed for the prediction of the sound absorption spectrum for one datum is on the scale of a millisecond (~1 s for 1000 absorbers), compared to almost 100 s with FEM, achieving a dramatic enhancement of 6 orders of magnitude.

Figure 3 demonstrates the range for each structural parameter in the training set $\mathbb{D}$ (solid line) and the testing set $\mathbb{D}_{\mathrm{I}}$ (blue and red triangles are for maximum and minimum, respectively). It is evident that $\mathbb{D}_{\mathrm{I}}$ contains a large number of structural configurations that $\overline{\mathbb{F}}$ has not been trained with (or "seen" metaphorically).

Figure 4 is the histogram of MSE of the 1000 sound absorption spectra predicted by $\overline{\mathbb{F}}$ for $\mathbb{D}_{\mathrm{I}}$. Overwhelming majority of the MSE values are under 0.01 and the maximum MSE is approximately 0.027, with the average MSE being roughly 1.6×10^{-3}. The outstanding prediction accuracy strongly demonstrates the forward network's ability in learning the hidden high dimensional non-linear relationship between the structure and the properties. In addition, the forward network enjoys high level of generalisation capability which enables it to be used for a wider range of "unseen" structural configurations without expanding the size of the training set.

4.2 Inverse network $\overline{\mathbb{I}}$

Generally during inverse designing, the 500 sound absorp-

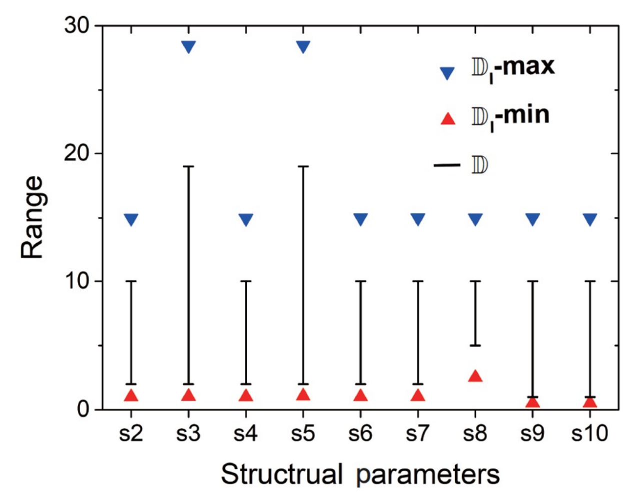

Figure 3 Comparison of the range of structural parameters (s2-s10) between the training set $\mathbb{D}$ and the testing set $\mathbb{D}_{\mathrm{I}}$.

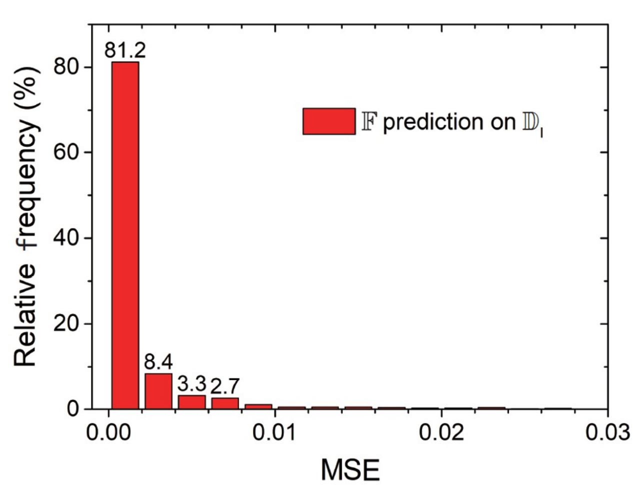

Figure 4 Statistics of MSE of the prediction on the sound absorption spectra for testing set I with $\overline{\mathbb{F}}$.

tion spectra in $\mathbb{D}_{\mathrm{II}}$ are input into $\mathbb{I}$ as the designing target, outputting 500 corresponding structural configurations whose acoustic properties best fit the targets the according to the "knowledge" of $\mathbb{I}$. Two experiments are carried out where "range1" (experiment1) and "range2" (experiment2) are applied to restrict the range of outputting structural parameters during inverse designing. Basically, this is to ask $\mathbb{I}$ to inversely design within the range it has "seen" (in the case of range1 for training set $\mathbb{D}$) or design more freely beyond the range considered in the training set.

Figure 5 is the statistics of the MSE in the sound absorption spectra comparing the target spectra and the spectra of the inversely designed structures. When spared the limitation of range1, the inverse design accuracy is tremendously improved, from only 9.2% of MSE values smaller than 0.0001 to 99.6%. The mean MSE decreases from 3.4×10^{-4} to 5.6×10^{-5}. Note that we are essentially asking $\mathbb{I}$ to inversely design for enhanced performance (i.e., the target acoustic properties surpass those in the training set). Therefore, $\mathbb{I}$ is delightfully capable of improving acoustic performance via spontaneously seeking beyond-range parameters.

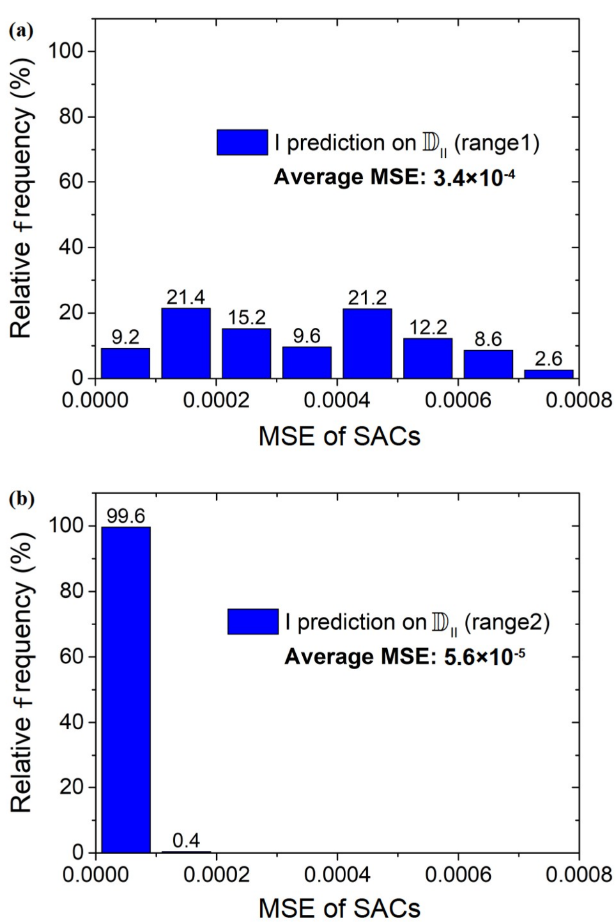

Figure 5 Statistics of the MSE of sound absorption spectra of the absorbers inversely designed by $\mathbb{I}$ with the target in the testing set II under the restriction of (a) range1 and (b) range2.

The minimum and maximum values for each inversely designed structural parameter are illustrated in Fig. 6. The black solid line with short horizontal lines at two ends is for the training set $\mathbb{D}$ while the blue and red triangles represent the maximum and the minimum values for the inversely designed structures. Small green arrows are placed to point out the parameters where the inversely designed parameters exceed the one in the training set $\mathbb{D}$. While moderate expansion exists in s2, s5, s7 and s10, s6 and s8 witness a fundamental shift in their values. Especially for s6, when $\mathbb{I}$ is allowed to inversely design beyond range1, it gives out completely different values, meaning that the model "knows" intelligently that an r_{22} value larger than all it has "seen" leads to a more accurate fit to the target. Therefore, the inverse model is incredibly capable of designing beyond the knowledge learned from the training set as long as it leads to a design that better meets the target.

4.3 Inverse design accuracy improvement thanks to its capability to go beyond-range

A case is chosen to demonstrate how $\mathbb{I}$ improves the inverse design accuracy via using beyond-range structural parameters. Fig. 7 is the comparison between the sound absorption spectrum target and the spectrum obtained via inverse design within range1 and range2, and they are respectively referred to as structure1 and structure2 in the following discussion. As depicted in Fig. 7, structure1 and structure2 exhibit similar acoustic behaviours in the low-frequency range though with rather different structural parameters. A large discrepancy exits from 4000 Hz to 8000 Hz for the range1 curve, leading to MSE value of 7.9×10^{-4}. As contrast, the structure2 spectrum achieves excellent agreement with the target across the whole frequency range

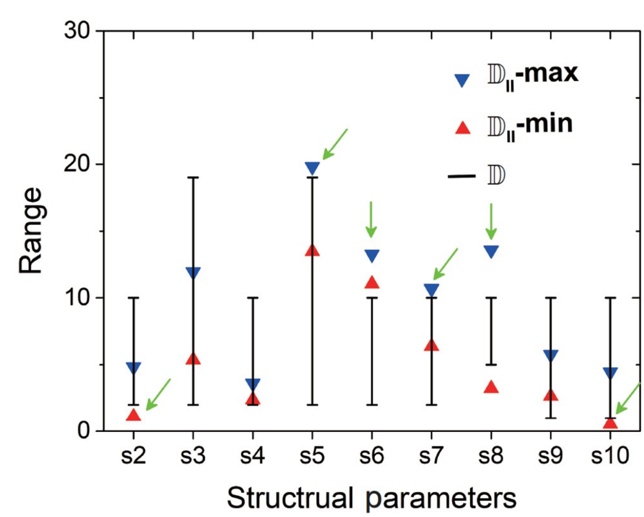

Figure 6 Range of structural parameters inversely designed by $\mathbb{I}$ with the spectra targets in testing set $\mathbb{D}_{\mathrm{II}}$.

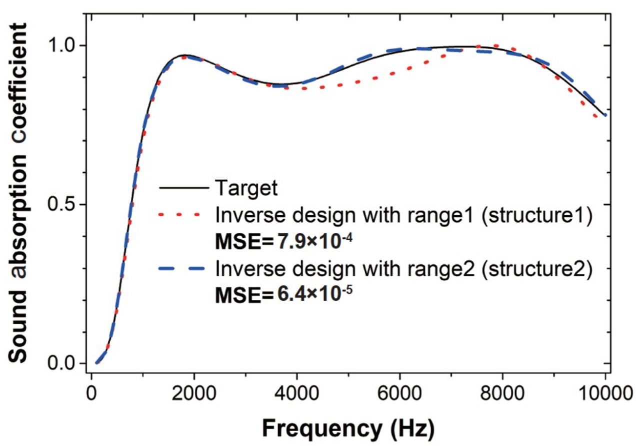

Figure 7 Comparison of the sound absorption spectra between the target from $\mathbb{D}_{\mathrm{II}}$ (black solid line) and the sound absorption spectra of inversely designed structures with range1 (structure1, red dotted line) and range2 (structure2, blue dashed line)

and the corresponding MSE is negligibly 6.4×10^{-5}, which is a huge improvement compared to the range1 case. The structural parameters of the two structures are listed in Table 4. Substantial difference of more than two times in the value is featured in s3 (h_t: the height of the cover plate), s8 (h_{c1}: the height of the cones), s9 (r_{11}: the radius of the upper base of the first cone) and s10 (r_{21}: the radius of the upper base of the second cone) while most of the other structural parameters experience noticeable but moderate changes.

Figure 8 is the acoustic pressure and displacement fields of structure1 and structure2 at 5000 Hz where the largest discrepancy exists in the sound absorption spectra between the two structures. In structure 1, the first absorption peak is induced by the beam-like resonance of the cover layers on the top of cavities 1 and 2, and the breathing resonance of the cavity 2. Whereas in structure2, owing to the small radii, the low-frequency resonance of the cavity 1 is suppressed, and the beam-like resonance of the cover layer on the top of cavity 2 and the breathing resonance of the cavity 2 dominate at the first absorption peak. It is known to us that the increment of the radius of the cavity and the decrement of the thickness of the cover layer would reduce the resonant frequency. Structure1 has a significantly larger upper-base radius (r_{11}) and a substantially thicker cover layer (h_t) compared to structure2, so the acoustic performance of them two at the first absorption peak may be similar with a certain combination of structural parameters. Beyond that, the large cavity radius would influence the acoustic impedance matching, leading to a low-efficiency absorption of structure1 in the high-frequency range. Whereas for structure2,

Table 4 Structural parameters in two experiments

	s1	s2	**s3**	s4	s5	s6	s7	**s8**	**s9**	**s10**
	a_0	r_{12}	$\boldsymbol{h_{c1}}$	r_{13}	h_{c2}	r_{22}	r_{23}	$\boldsymbol{h_t}$	$\boldsymbol{r_{11}}$	$\boldsymbol{r_{21}}$
structure1	97.79	2.02	**2.00**	2.0	18.99	10	10	**9.35**	**9.98**	**1.00**
structure2	99.17	1.11	**11.56**	3.25	14.35	13.27	9.89	**3.26**	**3.97**	**3.68**

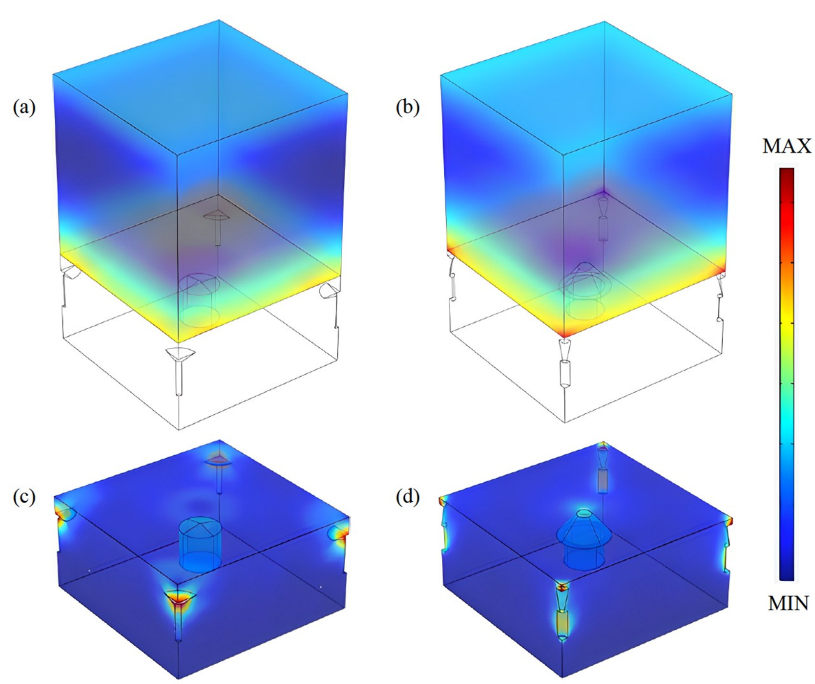

Figure 8 Acoustic pressure and displacement fields of structure1 and structure2 at 5000 Hz: (a) the pressure field of structure1; (b) the pressure field of structure2; (c) the displacement field of structure1; (d) the displacement field of structure2.

the cavity radius closer to the water domain is small and increases gradually along the incident direction, allowing the incident waves to adequately come into the matrix and contributing to a high-efficiency absorption in the high-frequency range.

With the increment of the lower-base radius (r_{22}) of the cone cavity 2, the first absorption peak moves to the lower frequencies with an increasing peak value, and correspondingly, the valley effect would be enhanced with a decreasing absorption performance following the absorption peak, due to the acoustic impedance mismatching, as shown in Fig. 9(a). Figure 9(b) illustrates that the absorption behavior in the high-frequency range can be improved by the addition of the height (h_t) of the cover plate with the low-frequency performance being slightly changed.

Additionally, an experiment is carried out to use the genetic algorithm as comparison to our inverse design approach. The aim of the genetic algorithm is to design a structure whose acoustic property matches the target sound absorption spectrum in Fig. 7 (black solid line). The genetic algorithm takes 64358 s (roughly 18 h) to reach the goal, while our method can complete 500 designs in roughly 4 s, achieving a huge improvement in efficiency.

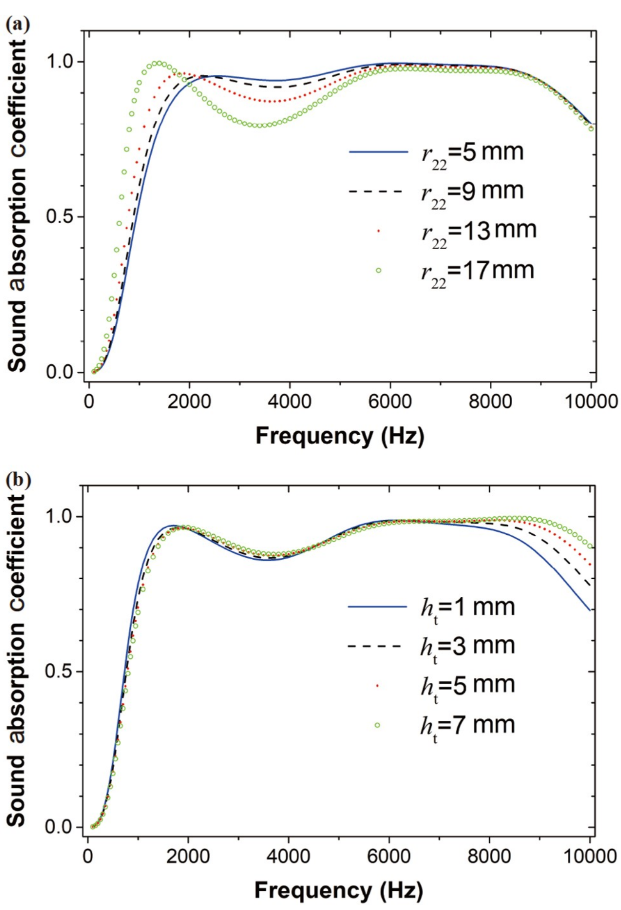

Figure 9 Influence of the lower-base radius (r_{22}) of the cone cavity in the middle (a) and the height (h_t) of the cover plate (b) on the absorption coefficient based on structure2.

5. Conclusion

An ANN-based inverse design framework consisting of a forward network and an inverse network was proposed in this work for the structure designing of 3D mixed-size cavity-based acoustic metamaterials for waterborne sound absorption under the target of on-demand broadband sound absorption spectra from 100 Hz to 10000 Hz. The training dataset, containing absorbers only with their mean SACs across the spectra smaller than 0.82, of structure-property pairs was prepared with COMSOL with a strict limit on the range (range1) for each structural parameter. The forward network is pre-trained to learn the hidden relationship between the structure and the acoustic properties while the inverse network is to output a structure that will meet the requirement of the sound absorption performance input as the target. Examining the forward network with testing dataset I of structures beyond range1, the prediction accuracy was exceptional with the average MSE negligibly being 1.6×10^{-3}, showing its excellent generalization. The forward network has improved the efficiency of acoustic properties evaluation of this type of absorbers by 6 orders of magnitude (from ~100 s to ~0.001 s). The trained inverse network took target-spectra from testing dataset II where absorbers all have >0.82 mean absorption coefficients and asked to output structures that feature matching spectra from within range1 to beyond. As a result, the inverse design accuracy is tremendously improved, from only 9.2% of MSE values smaller than 0.0001 to 99.6%. Therefore, the inverse framework is proven capable to design for enhanced acoustic performance by spontaneously adopting beyond-range structures beyond what is considered in the training set.

A case was chosen to demonstrate how the inverse model managed to superbly meet the acoustic property target by adopting structural parameters beyond range1 while the designing was far less satisfying with range1 restriction. This work, to the best of our knowledge, studies the capability of inverse designing with machine learning for enhance performance, which can be an inspiration for the design and optimization of elastic metamaterials. By opening up opportunities for new structure exploration for enhanced properties, more future work can be focused on including more designing degrees of freedom into the workflow to best stimulate its potential on producing higher-performance metamaterials.

Conflict of interest *On behalf of all authors, the corresponding author states that there is no conflict of interest.*

Author contributions *Hongjia Zhang designed and conducted the research and wrote the first draft of the manuscript. Jiawei Liu, Weitong Ma, Haitao Yang, and Yang Wang prepared the data. Haibin Yang, Honggang Zhao, Dianlong Yu, and Jihong Wen helped organize the manuscript and revised the final version. Hongjia Zhang, Honggang Zhao, and Jihong Wen provided the funding.*

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学习反向设计声学超材料以提高性能

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摘要 弹性超材料可被用于实现多种特殊功能, 例如振动控制和波操纵, 其中吸声是声学超材料完成的典型任务. 利用机器学习方法反向设计超材料因利用数据驱动且不依赖经验的优势而备受关注, 成为重要的设计范式之一. 但现有的工作大多集中在验证反向设计神经网络的设计准确性, 很少有人探索如何利用神经网络进行反向设计以提高材料性能. 为此, 我们的工作研究了反向设计框架在提升3D多尺度空腔型声学超材料的声学性能方面的能力. 框架中有正向和反向网络, 在训练过程中将目标吸声曲线(100~10000 Hz)作为反向网络的输入, 并输出具有满足此吸声曲线的相应超材料结构, 随后将其输入正向网络进行声学性能评估. 结果表明, 经过训练的前向网络可对超过训练集结构参数范围的结构(既"未见过"的结构)进行高精度性能预测, 因此具有较高泛化性能. 更重要的是, 反向网络能够自发地采用超出范围限制的结构参数, 以确保满足平均吸声系数高于训练集中任何数据的吸声目标, 因此可利用反向设计突破训练集中数据的声学性能, 进行性能优化. 通过超限探索, 反向设计精度从仅有9.2% 的设计拥有<0.0001的均方误差显着提高到 99.6%. 最后, 利用案例证明神经网络拥有的超限探索能力对旨在提高性能的反向设计有重要意义. 希望这项工作能为更高性能弹性超材料的优化设计提供支撑.